

PORTFOLIO RISK REVIEW

Credit Risk Analysis

Identify the characteristics of high-risk borrowers and evaluate lending strategies that improve portfolio quality.

255,347

LOANS

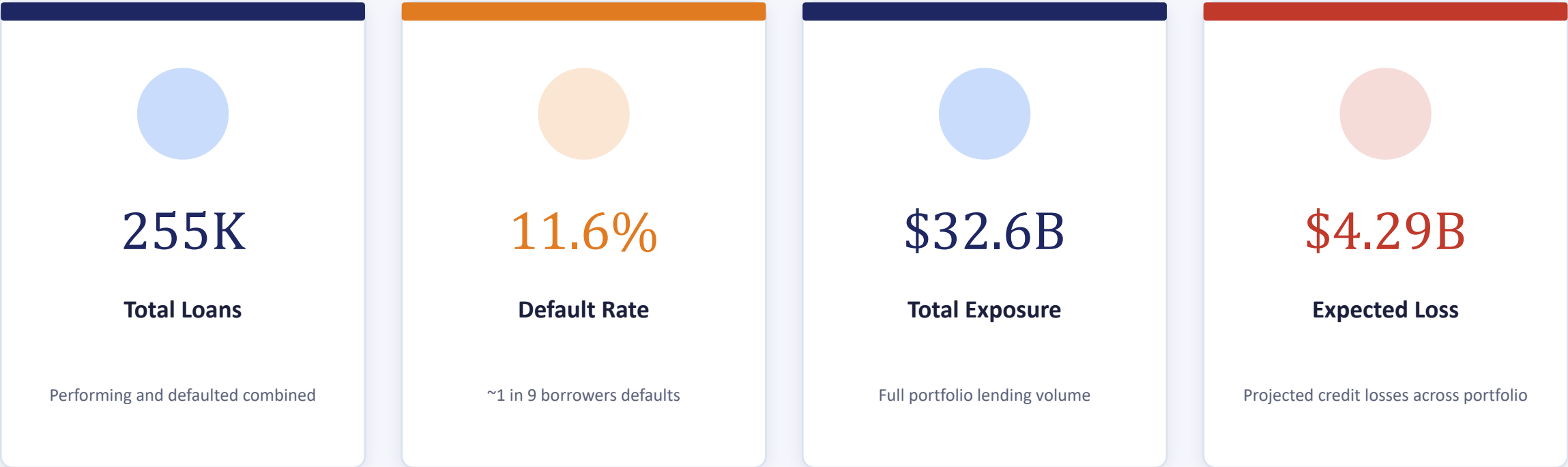
11.6%

DEFAULT RATE

\$32.6B

PORTFOLIO

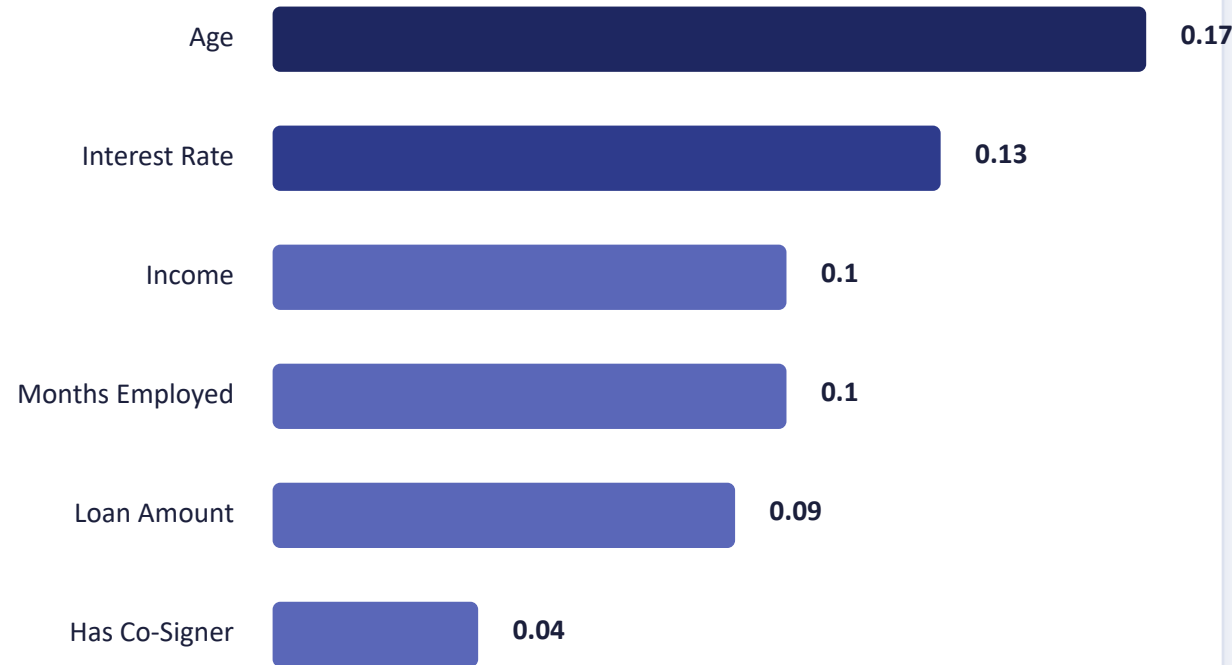
What Does Our Portfolio Look Like?



Reading the numbers: roughly 1 in 9 loans defaults, and the portfolio's expected loss represents about 13% of total exposure — the baseline this analysis aims to reduce.

What Drives Default Risk?

Absolute Correlation with Default



What the Data Tells Us

Age and Interest Rate are the strongest signals — older borrowers and lower rates correlate with lower default risk.

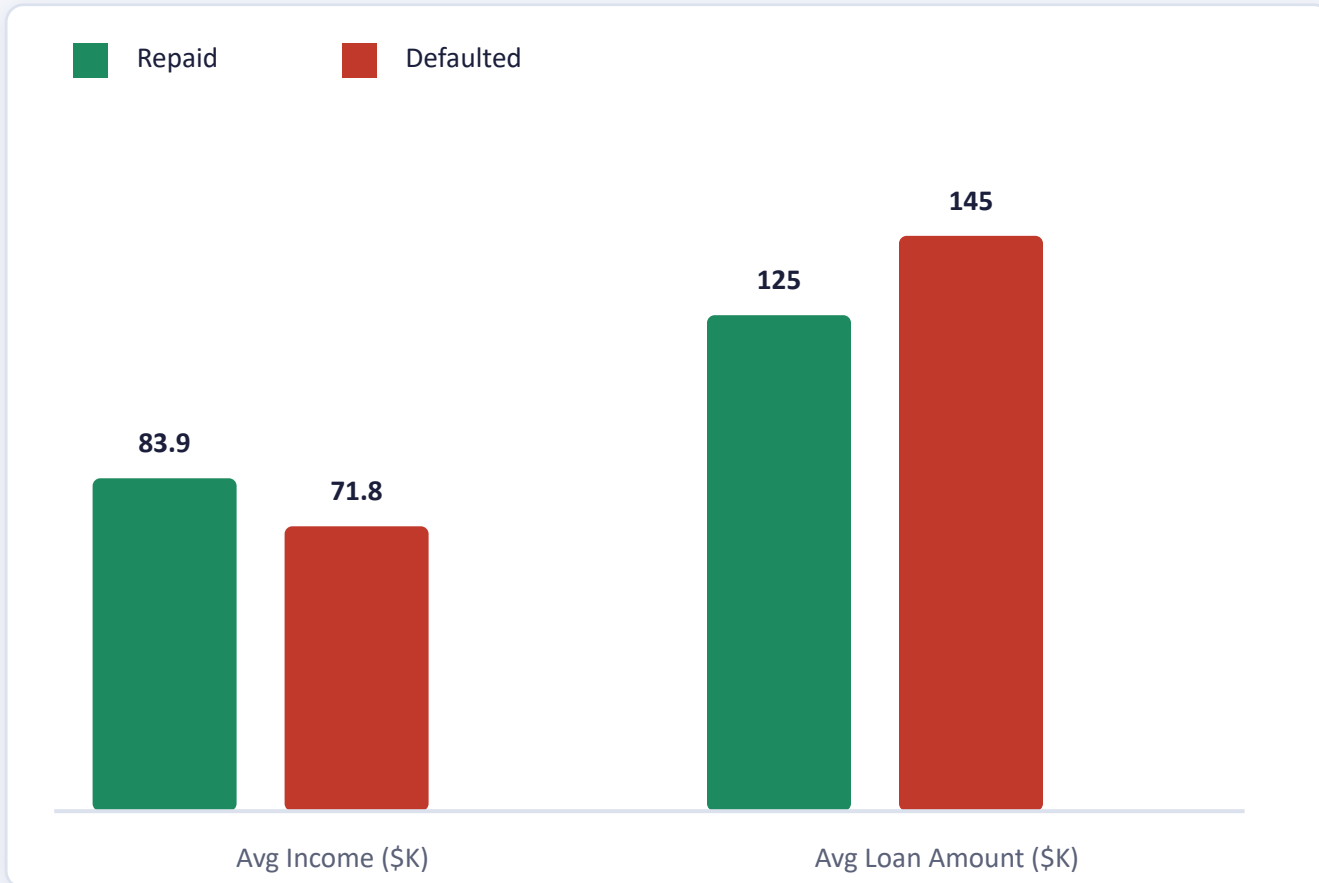
Income and employment tenure show moderate protective effects.

Credit Score has a weak standalone relationship (0.034) — not a dominant predictor in this portfolio.

No single variable strongly predicts default. Risk is multidimensional — it takes a combination of factors to identify true high-risk borrowers.

Who Are Our Defaulting Borrowers?

Defaulted borrowers share a consistent financial mismatch: **lower income, larger loans, higher repayment pressure.**

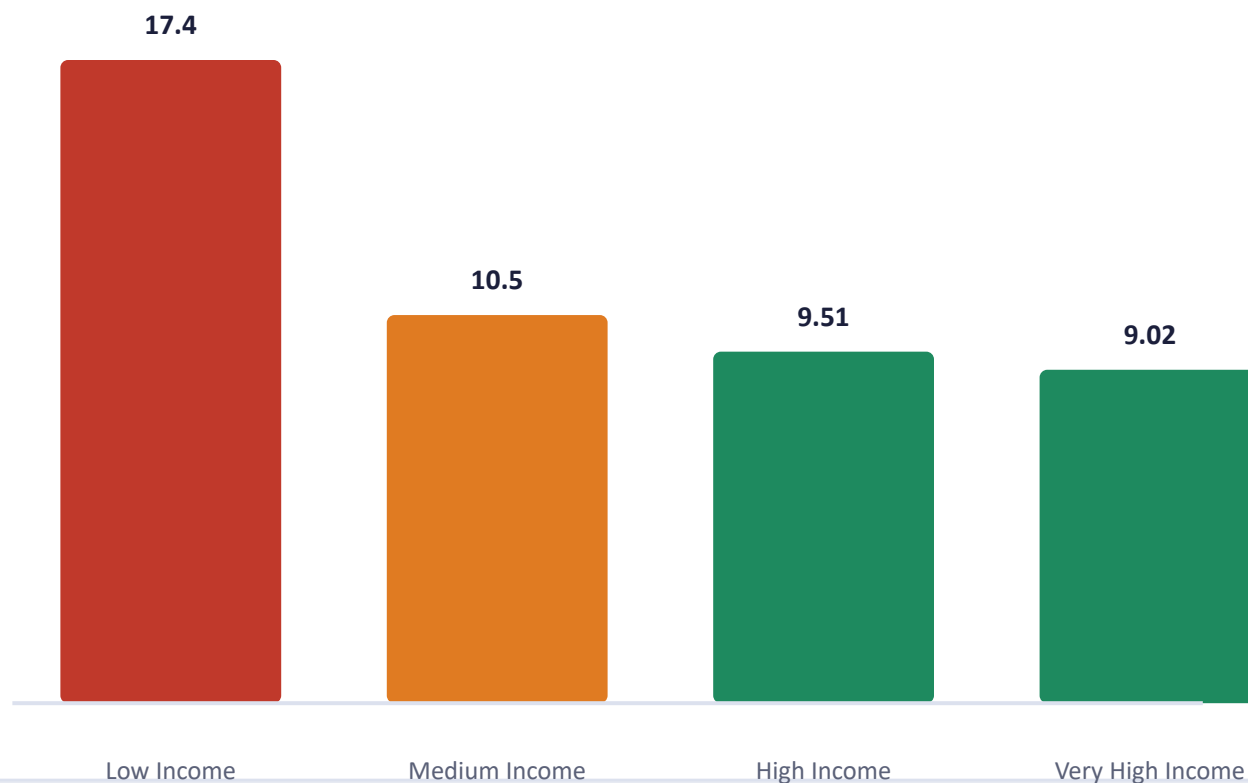


Key Differences

- **Income gap:** Defaulters earn \$12K less on average
- **Loan size:** Defaulters borrow \$19K more on average
- **Credit Score:** Slightly lower (559 vs 576) — not a strong differentiator
- **Employment:** Unemployed borrowers default at 13.6% vs 9.5% for full-time

Which Borrowers Are Highest Risk?

Default Rate by Income Segment (%)



Inverse Relationship: Income vs Default

Low-income borrowers default at **17.4%** — nearly **2× the rate** of high-income borrowers.

Loan amounts are **identical across all segments** (~\$127K), meaning loan size is not risk-adjusted by income.

Low-income segment: 63,837 borrowers · \$8.15B exposure · 17.66% predicted PD

How Well Does Our Model Perform?

Approval decision threshold: 0.2

True Positives

39,234

False Negatives

5,905

False Positives

3,307

Risky clients approved

True Negatives

2,624

Risky clients rejected

Portfolio Simulation Results

- **Approval rate:** 83% of applications approved
- **Risk in approved portfolio:** 7.8% (down from 11.6% baseline)
- **Net simulated profit:** \$166.4M
- **ROI:** 2.55% on test set exposure
- **ROC-AUC:** 0.751 · Accuracy: 88.6%

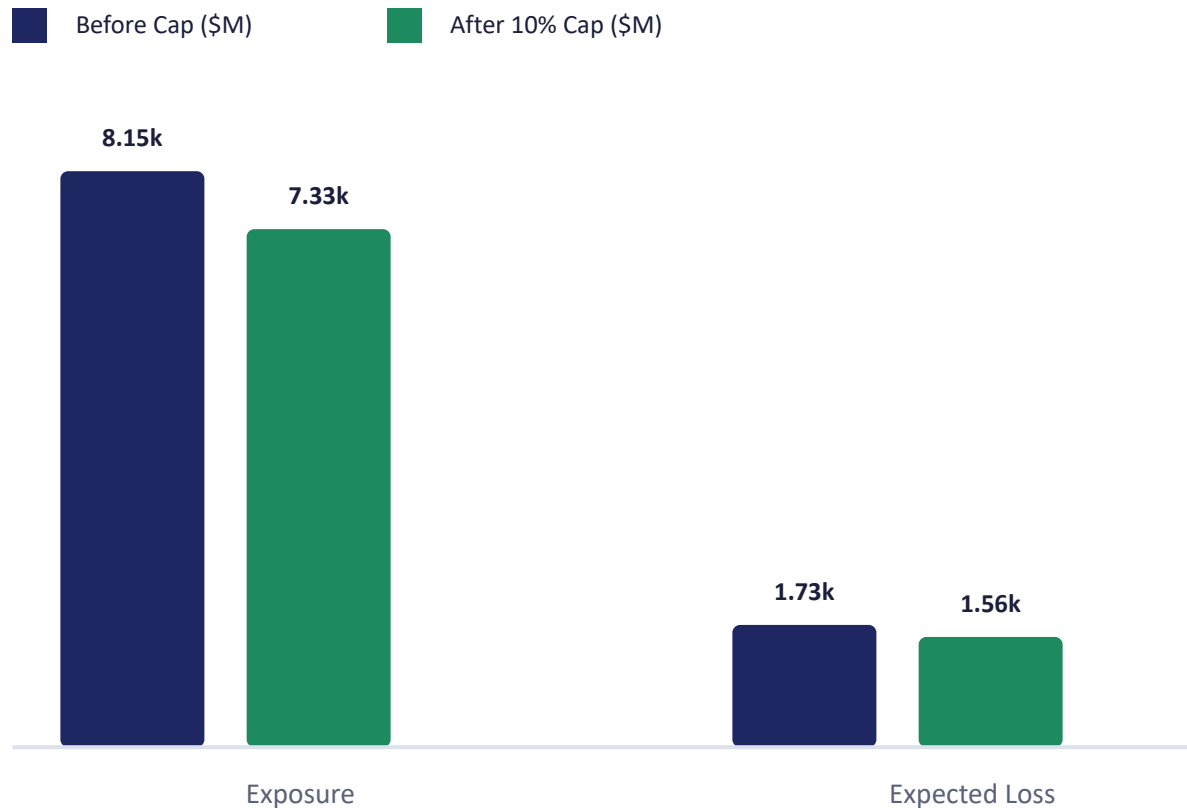
Model detects only 3% of defaulters — highly conservative in risk flagging, maximizing volume and profit over aggressive risk capture.

1

STRATEGY 1

Loan Cap on Low-Income Borrowers

10% Loan Cap — Low-Income Segment



Impact of the Cap

- **Exposure reduction:** \$814.7M
- **Expected loss reduction:** \$173.2M
- **Efficiency:** 21.3 cents saved per \$1 of lending removed
- **Avg loan:** \$127,624 → \$114,862 per borrower

Loan amounts are uniform across income groups — capping low-income borrowers directly reduces concentrated risk without restructuring the portfolio.

Targeted High-Risk Segment Cap

Ultra-targeted filter: **Low Income + Credit Score < 350 + DTI > 0.60** → 2,109 borrowers, **20.6% observed default rate**

10% Cap

Exposure reduced

\$26.8M

Loss reduced

\$6.7M

Efficiency

25.1%

20% Cap

Exposure reduced

\$53.7M

Loss reduced

\$13.5M

Efficiency

25.1%

Combined Hybrid

Total exposure reduced

~\$868M

Total loss reduced

~\$179M

Efficiency

21–25%

3

STRATEGY 3

High-Risk Borrower Rejection Policy

Reject applications meeting all three criteria: **Credit Score < 400** · **Bottom 25% Income** · **DTI > 55%**

Expected Loss (\$M)

4.29k



4.13k



Before Policy

After Policy

Portfolio Impact

- **5,071 applications** rejected — small fraction of portfolio
- **\$645.6M** lending volume removed
- **\$159.7M** expected loss avoided
- **3.73% reduction** in total expected portfolio loss
- **Default rate:** 11.61% → 11.44% (–0.17 pp)

A narrow, precisely targeted policy — rejecting the highest-risk 2% of applications avoids a disproportionate share of expected losses.

THE RECOMMENDATION

Recommended Strategy: Hybrid Approach

1

Apply 10% Loan Cap — All Low-Income Borrowers

Reduces \$814.7M exposure and \$173.2M expected loss across 63,837 borrowers

2

Apply 20% Loan Cap — Ultra High-Risk Segment

Targets Credit Score < 350 + DTI > 0.60 — additional \$53.7M exposure and \$13.5M loss reduction

3

Reject: Score < 400 + Low Income + DTI > 55%

Eliminates \$159.7M in expected losses by declining only 5,071 highest-risk applications

Combined impact: ~\$868M exposure reduction · ~\$179–\$339M expected loss reduction · 21–25% efficiency